

Unified vehicle calibration report

Session: `real_steering_20260625`

This document is regenerated after every completed stage. Each following page records the capture, analysis/update, and independent-validation role of one stage.

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Steering command-chain audit

Stage key: steering_command_audit

Status: pending

A — Capture

A — Record the stage-specific data.

B — Analysis and candidate update

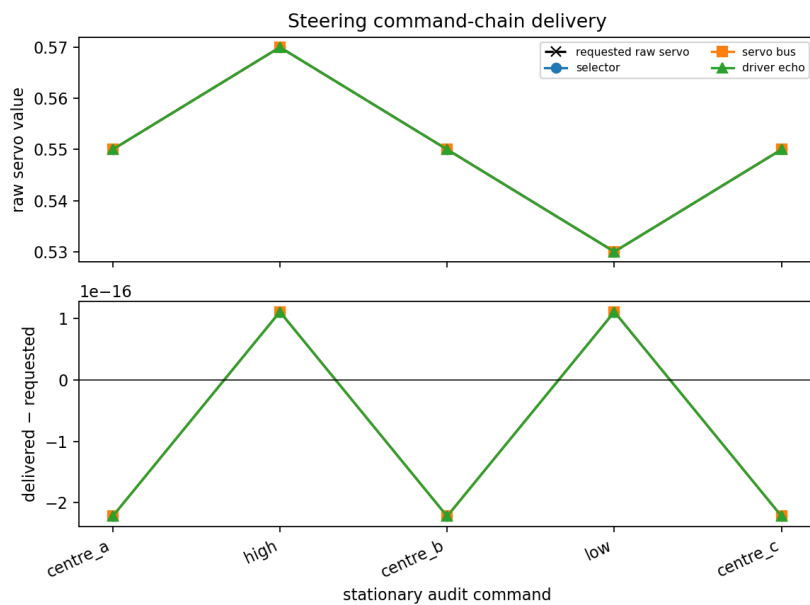
B — Run its quality analysis and preserve the result.

C — Independent validation

C — Use the result as a prerequisite or independent validation for downstream fitted stages.

Measured / fitted values

- No scalar result has been written yet.



Stage plot: Steering command-chain audit

Raw data: ../00_command_chain_audit/

Stage report: stage_reports/steering_command_audit.md

Direct physical vehicle metrology

Stage key: physical_metrology

Status: pending

A — Capture

A — Measure the race-ready car using the session-local physical-measurements sheet.

B — Analysis and candidate update

B — Check SI units and mass/axle-load closure; derive l_f/l_r , freeze LiDAR/rear-axle/IMU reference geometry, apply the reversible dynamic-model geometry patch, and rebuild.

C — Independent validation

C — This is direct metrology, not a fitted time-series model; later independent motion stages use the rebuilt geometry and it remains an audited prerequisite for lateral identification.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../00a_physical_metrology/

Stage report: stage_reports/physical_metrology.md

LiDAR/IMU observability check

Stage key: steering_observability

Status: pending

A — Capture

A — Record the stage-specific data.

B — Analysis and candidate update

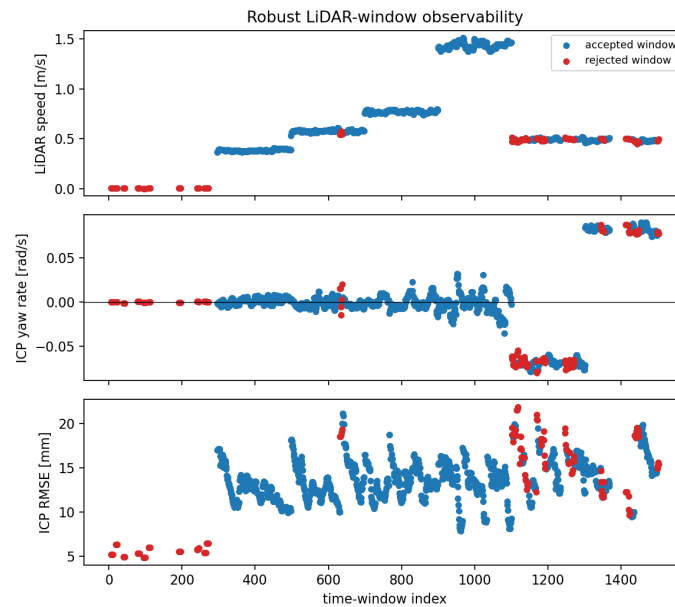
B — Run its quality analysis and preserve the result.

C — Independent validation

C — Use the result as a prerequisite or independent validation for downstream fitted stages.

Measured / fitted values

- No scalar result has been written yet.



Stage plot: LiDAR/IMU observability check

Raw data: ../03_sensor_observability/

Stage report: stage_reports/steering_observability.md

Steering offset: training capture and fit

Stage key: steering_centre

Status: pending

A — Capture

A — Record independent training captures.

B — Analysis and candidate update

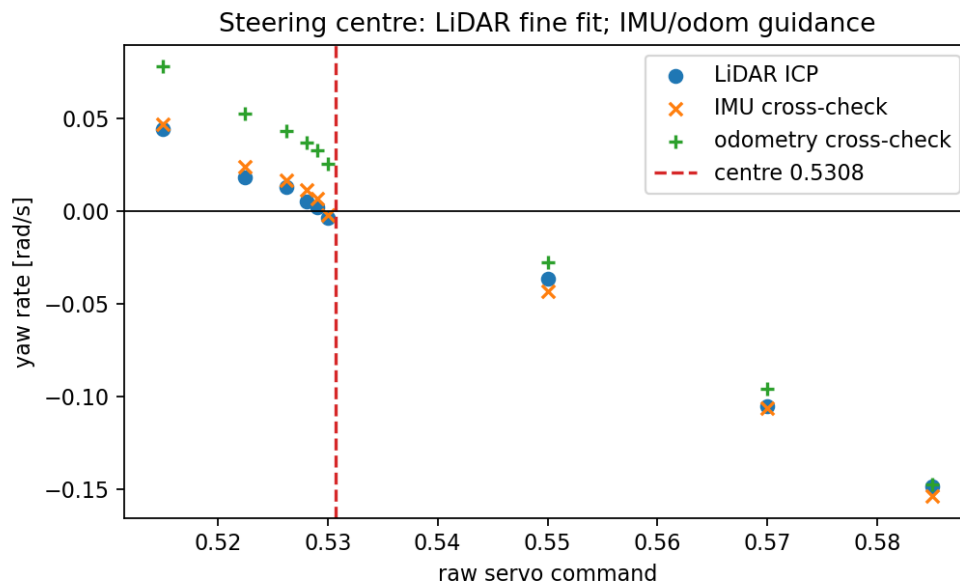
B — Fit, gate, apply the candidate patch, and rebuild inside the reversible transaction.

C — Independent validation

C — The following hold-out/validation stage evaluates the applied candidate on new data.

Measured / fitted values

- No scalar result has been written yet.



Stage plot: Steering offset: training capture and fit

Raw data: ../01_zero_curvature_centre/

Stage report: stage_reports/steering_centre.md

Steering offset: independent straightness validation

Stage key: steering_centre_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../01a_zero_curvature_validation/

Stage report: stage_reports/steering_centre_validation.md

Human steering-limit survey

Stage key: steering_endstops

Status: pending

A — Capture

A — Human records the last mechanically free steering limit on each side.

B — Analysis and candidate update

B — The safe raw-servo envelope is stored and checked against the validated centre.

C — Independent validation

C — This is a manual safety prerequisite, not a statistical fitted model.

Measured / fitted values

- No scalar result has been written yet.



Safe low: 0.1160
Centre: 0.5325
Safe high: 0.9340
Servo safety margin: 0.006

Stage plot: Human steering-limit survey

Raw data: ../02_physical_endstops/

Stage report: stage_reports/steering_endstops.md

Servo-to-effective-steering training

Stage key: steering_static_training

Status: pending

A — Capture

A — Record independent training captures.

B — Analysis and candidate update

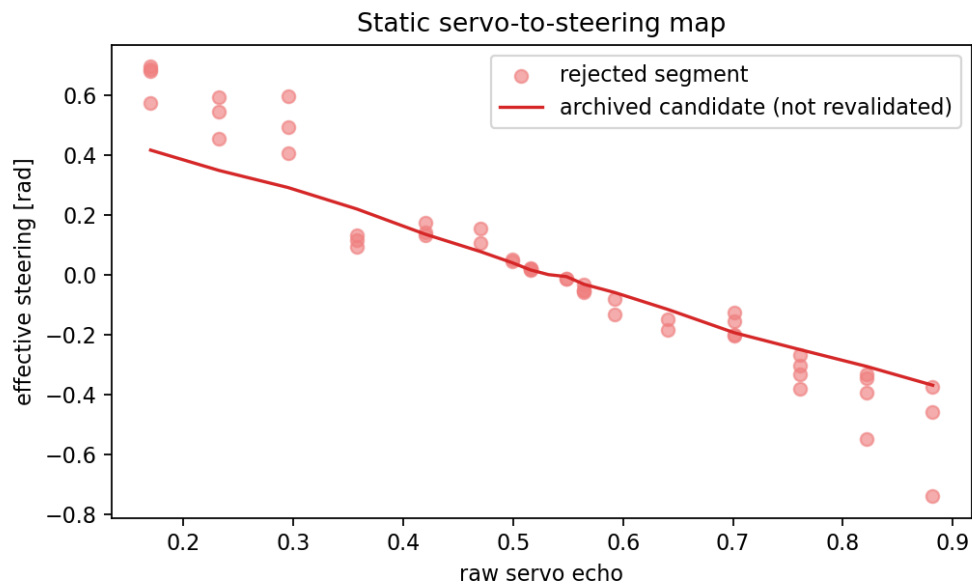
B — Fit, gate, apply the candidate patch, and rebuild inside the reversible transaction.

C — Independent validation

C — The following hold-out/validation stage evaluates the applied candidate on new data.

Measured / fitted values

- No scalar result has been written yet.



Stage plot: Servo-to-effective-steering training

Raw data: ../04_static_map_training/

Stage report: stage_reports/steering_static_training.md

Servo-to-effective-steering hold-out validation

Stage key: steering_static_holdout

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

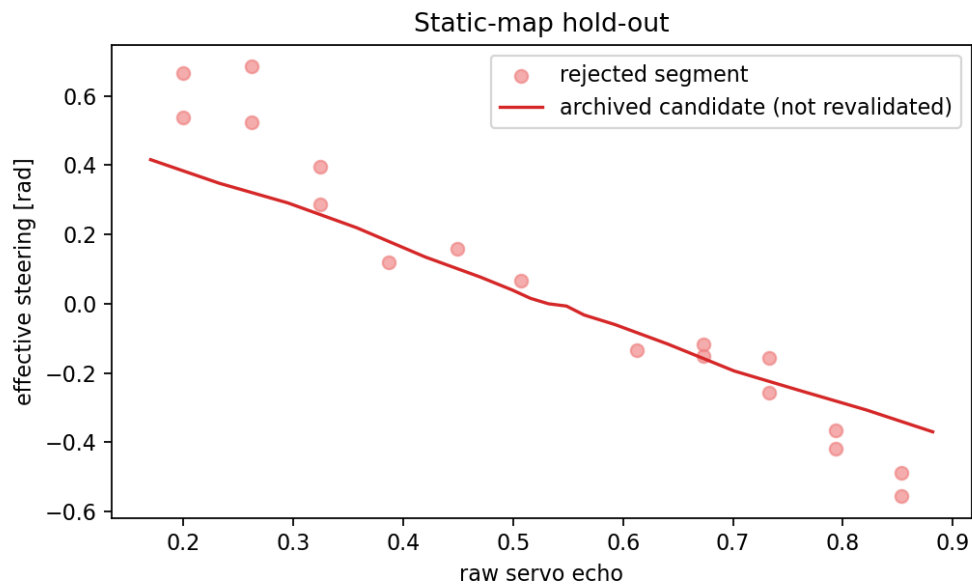
B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.



Stage plot: Servo-to-effective-steering hold-out validation

Raw data: ../05_static_map_holdout/

Stage report: stage_reports/steering_static_holdout.md

Steering response: training characterisation

Stage key: steering_response

Status: pending

A — Capture

A — Record the stage-specific candidate/training command grid.

B — Analysis and candidate update

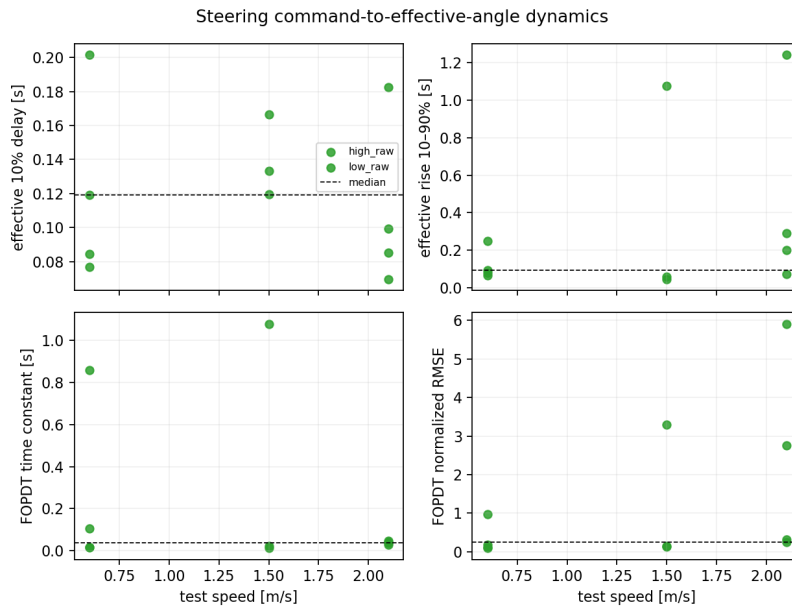
B — Estimate timing/routing metrics and preserve the candidate evidence; no hidden refit is made on later C data.

C — Independent validation

C — The following explicitly named hold-out stage records different commands and evaluates the frozen result.

Measured / fitted values

- No scalar result has been written yet.



Stage plot: Steering response: training characterisation

Raw data: ../06_command_to_curvature_response/

Stage report: stage_reports/steering_response.md

Steering response: independent hold-out

Stage key: steering_response_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../06a_command_to_curvature_response_validation/

Stage report: stage_reports/steering_response_validation.md

Motor command-chain audit

Stage key: motor_command_audit

Status: pending

A — Capture

A — Record the stage-specific data.

B — Analysis and candidate update

B — Run its quality analysis and preserve the result.

C — Independent validation

C — Use the result as a prerequisite or independent validation for downstream fitted stages.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../00b_motor_command_chain_audit/

Stage report: stage_reports/motor_command_audit.md

Longitudinal sensor observability

Stage key: longitudinal_observability

Status: pending

A — Capture

A — Capture a stationary diagnostic and repeated low/medium/high-speed straight passes.

B — Analysis and candidate update

B — Gate the robust moving LiDAR-window measurement quality and write the stationary noise diagnostic.

C — Independent validation

C — Later ERPM training is blocked unless this independent moving-sensor preflight passed.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../01_longitudinal_observability/

Stage report: stage_reports/longitudinal_observability.md

Low-speed launch characterisation

Stage key: low_speed_launch

Status: pending

A — Capture

A — Capture repeated raw-ERPM launches at the low-speed grid.

B — Analysis and candidate update

B — Report condition coverage and the first repeatable LiDAR ground-motion threshold; do not prematurely fit the full speed map.

C — Independent validation

C — The later rebuilt VEL_TO_ERPM pipeline audit checks the resulting slow-start setting on new commands.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../02_low_speed_launch/

Stage report: stage_reports/low_speed_launch.md

ERPM-to-ground-speed training

Stage key: erpm_map_training

Status: pending

A — Capture

A — Record independent training captures.

B — Analysis and candidate update

B — Fit, gate, apply the candidate patch, and rebuild inside the reversible transaction.

C — Independent validation

C — The following hold-out/validation stage evaluates the applied candidate on new data.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../03_raw_erpm_map_training/

Stage report: stage_reports/erpm_map_training.md

ERPM-to-ground-speed hold-out validation

Stage key: erpm_map_holdout

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../04_raw_erpm_map_holdout/

Stage report: stage_reports/erpm_map_holdout.md

Velocity-to-ERPM command-path validation

Stage key: `vel_to_erpm_audit`

Status: pending

A — Capture

A — Record the stage-specific data.

B — Analysis and candidate update

B — Run its quality analysis and preserve the result.

C — Independent validation

C — Use the result as a prerequisite or independent validation for downstream fitted stages.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: `../05_vel_to_erpm_pipeline_audit/`

Stage report: `stage_reports/vel_to_erpm_audit.md`

ERPM response characterisation

Stage key: erpm_response

Status: pending

A — Capture

A — Record the stage-specific candidate/training command grid.

B — Analysis and candidate update

B — Estimate timing/routing metrics and preserve the candidate evidence; no hidden refit is made on later C data.

C — Independent validation

C — The following explicitly named hold-out stage records different commands and evaluates the frozen result.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../06_raw_erpm_response/

Stage report: stage_reports/erpm_response.md

ERPM response independent hold-out

Stage key: erpm_response_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../06a_raw_erpm_response_validation/

Stage report: stage_reports/erpm_response_validation.md

Coast-down drag fit

Stage key: coastdown

Status: pending

A — Capture

A — Record independent training captures.

B — Analysis and candidate update

B — Fit, gate, apply the candidate patch, and rebuild inside the reversible transaction.

C — Independent validation

C — The following hold-out/validation stage evaluates the applied candidate on new data.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../07_coastdown/

Stage report: stage_reports/coastdown.md

Coast-down drag independent hold-out

Stage key: coastdown_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../07a_coastdown_validation/

Stage report: stage_reports/coastdown_validation.md

Current-to-acceleration training

Stage key: `current_training`

Status: `pending`

A — Capture

A — Record independent training captures.

B — Analysis and candidate update

B — Fit, gate, apply the candidate patch, and rebuild inside the reversible transaction.

C — Independent validation

C — The following hold-out/validation stage evaluates the applied candidate on new data.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: `../08_raw_current_training/`

Stage report: `stage_reports/current_training.md`

Current-to-acceleration hold-out validation

Stage key: `current_holdout`

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: `../09_raw_current_holdout/`

Stage report: `stage_reports/current_holdout.md`

Acceleration-command interface training audit

Stage key: accel_interface

Status: pending

A — Capture

A — Record the stage-specific candidate/training command grid.

B — Analysis and candidate update

B — Estimate timing/routing metrics and preserve the candidate evidence; no hidden refit is made on later C data.

C — Independent validation

C — The following explicitly named hold-out stage records different commands and evaluates the frozen result.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../10_accel_to_current_interface/
Stage report: stage_reports/accel_interface.md

Acceleration-command interface independent hold-out

Stage key: accel_interface_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../10a_accel_to_current_interface_validation/
Stage report: stage_reports/accel_interface_validation.md

Selected odometry model: fresh steady-speed validation

Stage key: odometry_candidate_velocity_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../11_candidate_velocity_verification/

Stage report: stage_reports/odometry_candidate_velocity_validation.md

Selected odometry model: fresh transient validation

Stage key: odometry_candidate_accel_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../12_candidate_accel_verification/

Stage report: stage_reports/odometry_candidate_accel_validation.md

Effective tyre stiffness: quasi-steady training arcs

Stage key: lateral_stiffness_training

Status: pending

A — Capture

A — Record balanced, quasi-steady left/right speed-and-steering arcs after physical metrology.

B — Analysis and candidate update

B — Fit effective low-slip front/rear stiffness plus the dynamic-model steering scale, apply the reversible bicycle-model patch, and retain a diagnostic nonlinear candidate.

C — Independent validation

C — The following hold-out uses distinct arcs and blocks promotion if the frozen scale/stiffness model fails or a nonlinear tyre model is materially better.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../12_quasi_steady_lateral_training/

Stage report: stage_reports/lateral_stiffness_training.md

Effective tyre stiffness: independent hold-out arcs

Stage key: lateral_stiffness_validation

Status: pending

A — Capture

A — Record a fresh validation data set after the preceding candidate was applied.

B — Analysis and candidate update

B — Analyse against the already fitted candidate; do not refit from the hold-out.

C — Independent validation

C — Pass only when the new-data gates and operator acceptance agree.

Measured / fitted values

- No scalar result has been written yet.

No plot has been generated for this stage yet.

Raw data: ../12a_quasi_steady_lateral_validation/

Stage report: stage_reports/lateral_stiffness_validation.md